

Complex numbers geometrically, and loci in the complex plane [Cambridge revision]

Modulus is distance, argument is direction — and every locus becomes a picture.

LESSON 95–96

2 × 40 MIN

ALGEBRA 11, QO‘SHIMCHA BO‘LIM

P2/P3 11.5–11.6

LESSON CLOCK

15'

25'

25'

25'

10'

Distance and direction	15 min
The three standard loci	25 min
Regions	25 min
Greatest and least	25 min
Homework	10 min

LEARNING OBJECTIVES

- Interpret $|z - a|$ as the distance from z to a .
- Recognise the circle, the perpendicular bisector and the half-line as loci.
- Sketch a region defined by inequalities in the complex plane.
- Find the greatest and least value of $|z|$ on a given locus.

TEXTBOOK

National	pp. 429–440
Cambridge	Pure Mathematics 2 & 3, pp. 263–276
Workbook	Exercise 11E–11F

01

Explanation

Distance and direction

TWO READINGS THAT UNLOCK EVERYTHING

$|z - a|$ is the **distance** from the point z to the point a

$arg(z - a)$ is the **direction** of z as seen from a

With $a = 0$ these become the modulus and argument of z itself.

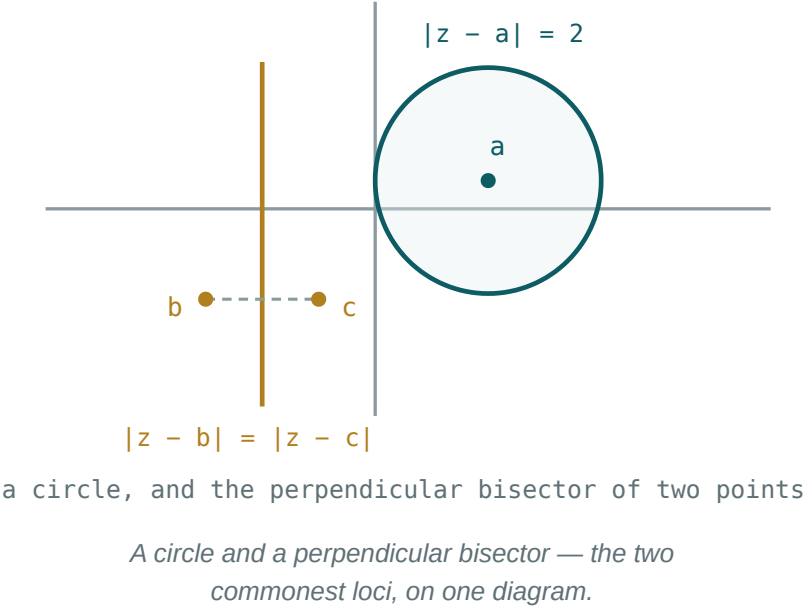
So a condition on $|z - a|$ is a condition on distance, and a condition on $arg(z - a)$ is a condition on direction. Every locus question in this lesson is one of the two.

$|z - A|$, NOT $|z + A|$

$|z + 3|$ is the distance to -3 , not to 3 — write it as $|z - (-3)|$ before reading off the centre. This single sign is where most locus marks are lost.

The three standard loci

CONDITION	LOCUS	DESCRIBED BY
$ z - a = r$	a circle	centre a , radius r
$ z - a = z - b $	a straight line	the perpendicular bisector of ab
$\arg(z - a) = \theta$	a half-line	from a (excluded) at angle θ



Worked readings.

- $|z - 2 - i| = 3$: circle, centre $2 + i$, radius 3.
- $|z| = |z - 4|$: the perpendicular bisector of 0 and 4 , that is the vertical line $\operatorname{Re}(z) = 2$.
- $\arg(z - 1) = \frac{\pi}{4}$: the half-line from 1 going up and to the right at 45° , with 1 itself excluded (the argument of 0 is undefined).

THE HALF-LINE IS HALF A LINE

$\arg(z - a) = \frac{\pi}{4}$ gives only the ray in that direction. The opposite ray has argument $-\frac{3\pi}{4}$, and belongs to a different locus. Drawing the whole line is the standard error.

Regions

Replace = by an inequality and the curve becomes a region — shaded, with the boundary solid or dashed exactly as on a number line.

CONDITION	REGION
$ z - a < r$	inside the circle, boundary excluded
$ z - a \geq r$	outside, boundary included
$ z - a \leq z - b $	the half-plane nearer a
$0 \leq \arg(z - a) \leq \frac{\pi}{2}$	a quarter-plane wedge at a

Two or three conditions together give the intersection — a lens, a segment of a disc, a wedge cut by a circle. Sketch each boundary first, then shade only where all conditions hold.

TEST ONE POINT

After shading, pick an easy point — often 0 — and check it satisfies every condition. If it does and it is inside your shading, the region is right; if it does and it is outside, you have shaded the complement.

Greatest and least

“Find the greatest value of $|z|$ on this locus” asks for the point of the locus furthest from the origin. On a circle this needs no calculus at all.

$|a| - r \leq |z| \leq |a| + r$ for z on the circle $|z - a| = r$

Both extremes are on the line through O and the centre: the nearest point on the near side, the furthest diametrically opposite.

LOCUS	$ A $	LEAST $ z $	GREATEST $ z $
$ z - 3 - 4i = 2$	5	3	7
$ z - 6i = 2$	6	4	8
$ z - 1 = 3$	1	0	4

The last row: the circle passes through the origin, so the least modulus is 0 — and $|a| - r$ would have given -2 , which is impossible. When the origin is inside or on the circle, the least value is 0 .

THE TRIANGLE INEQUALITY BEHIND IT

$$||z| - |a|| \leq |z - a| \leq |z| + |a|$$

Both bounds above are this inequality with $|z - a| = r$, and both are attained — which is what makes the answers exact rather than merely bounds.

02 Worked examples

EXAMPLE 1 Describe the locus $|z - 2 - i| = 3$ and sketch it.

- 1 Write it as $|z - (2 + i)| = 3$.
- 2 Distance from z to $2 + i$ is 3.
- 3 A circle, centre $(2, 1)$, radius 3.

ANSWER

A circle of centre $2 + i$ and radius 3

EXAMPLE 2 Describe the locus $|z - 1| = |z + 3|$.

- 1 $|z - 1| = |z - (-3)|$
Both as distances.
- 2 Equidistant from 1 and -3 .
- 3 The perpendicular bisector: the vertical line $\operatorname{Re}(z) = -1$.

ANSWER

The line $\operatorname{Re}(z) = -1$

EXAMPLE 3 Find the greatest and least values of $|z|$ for $|z - 3 - 4i| = 2$.

- ① $|a| = |3 + 4i| = 5$
- ② Least $5 - 2 = 3$.
- ③ Greatest $5 + 2 = 7$.
Both on the line through the origin and the centre.

ANSWER

3 and 7

EXAMPLE 4 Sketch the region $|z - 2i| \leq 2$ and $0 \leq \arg z \leq \frac{\pi}{2}$.

- ① Disc of centre $2i$, radius 2, boundary included.
- ② Wedge from the origin between the positive real and positive imaginary axes.
- ③ The intersection is the right half of the disc.
- ④ Check $z = 1 + 2i$: in the disc, and $\arg \approx 63^\circ$ — inside.

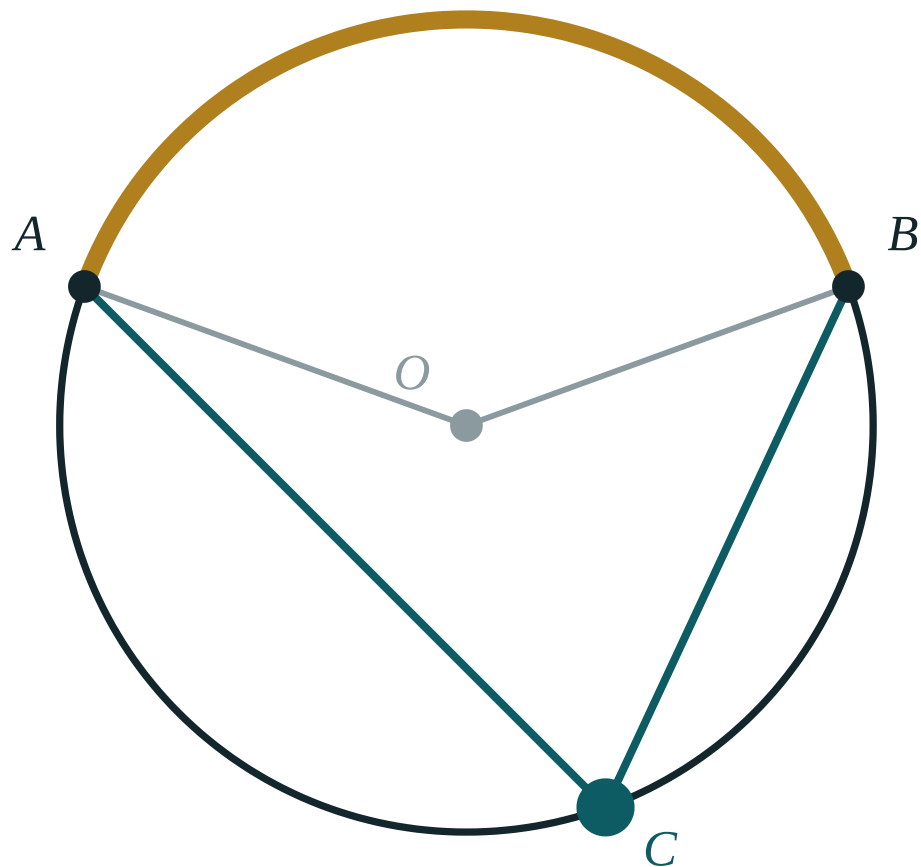
ANSWER

The right half of the disc, boundaries included

03 Interactive model

Give each pair squared paper and four loci to draw before any algebra; the descriptions then write themselves.

Circles, chords and directions

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Locus	Geometrik o'rin	Геометрическое место
Distance	Masofa	Расстояние
Circle	Aylana	Окружность
Perpendicular bisector	O'rta perpendikulyar	Серединный перпендикуляр
Half-line (ray)	Nur	Луч
Region	Soha	Область
Inequality	Tengsizlik	Неравенство
Greatest value	Eng katta qiymat	Наибольшее значение
Least value	Eng kichik qiymat	Наименьшее значение
Triangle inequality	Uchburchak tengsizligi	Неравенство треугольника

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$|z - a|$ means:

the argument

the distance from z to a

the sum

the product

$|z - a| = r$ is:

a line

a circle

a half-line

a point

$|z - a| = |z - b|$ is:

a circle

the perpendicular bisector of ab

a half-line

a point

$\arg(z - a) = \theta$ is:

a whole line

a half-line from a

a circle

a wedge

$|z + 3|$ is the distance to:

3

-3

$3i$

0

On $|z - 3 - 4i| = 2$, the greatest $|z|$ is:

3

5

7

9

If the origin is inside the circle, the least $|z|$ is:

$|a| - r$

0

r

$|a|$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Describe $|z| = 4$

ANSWER Circle, centre 0 , radius 4

2. Describe $|z - 2| = 3$

ANSWER Circle, centre 2 , radius 3

3. Describe $|z + 3| = 1$

ANSWER Circle, centre -3 , radius 1

4. Describe $|z| = |z - 4|$

ANSWER The line $\operatorname{Re}(z) = 2$

5. Describe $\arg z = \frac{\pi}{4}$

ANSWER Half-line from 0 at 45°

6. Describe $|z - i| < 2$

ANSWER Inside the circle, boundary excluded

7. Centre of $|z - 2 - i| = 3$

ANSWER $2 + i$

Medium · the standard question

7 PROBLEMS

1. Describe $|z - 1| = |z + 3|$

ANSWER The line $\operatorname{Re}(z) = -1$

2. Describe $|z - 2i| = |z - 4|$

ANSWER The perpendicular bisector of $2i$ and 4

3. Describe $\arg(z - 1) = \frac{\pi}{4}$

ANSWER Half-line from 1 , excluded

4. Greatest $|z|$ on $|z - 3 - 4i| = 2$

ANSWER 7

5. Least $|z|$ on the same

ANSWER 3

6. Least $|z|$ on $|z - 1| = 3$

ANSWER 0

7. Describe $|z - 2| \leq |z|$

ANSWER The half-plane $\operatorname{Re}(z) \geq 1$

Hard · stretch and reason

7 PROBLEMS

1. Greatest and least $|z - 1|$ on $|z - 5 - 3i| = 2$

ANSWER 7 and 3

2. Sketch $|z - 2i| \leq 2$ and $0 \leq \arg z \leq \frac{\pi}{2}$

ANSWER The right half of the disc

3. Describe $|z - 1| = 2|z + 1|$

ANSWER A circle — centre $-\frac{5}{3}$, radius $\frac{4}{3}$

4. Greatest $\arg z$ on $|z - 4i| = 2$

ANSWER $\frac{2\pi}{3}$ — at the tangent from O

5. The region $|z| \leq 3$ and $|z - 3| \leq 3$: describe it

ANSWER A lens between the two circles

6. Show that $|z_1 + z_2| \leq |z_1| + |z_2|$ with equality when the arguments are equal

ANSWER The triangle inequality

7. Find the points on $|z - 3 - 4i| = 2$ of greatest and least modulus

ANSWER $\frac{21 + 28i}{5}$ and $\frac{9 + 12i}{5}$

07 Homework

Homework — 5 tasks

Draw every locus before describing it; a sketch earns marks that words alone do not.

- Describe and sketch $|z - 3 + 2i| = 4$, giving the centre and the radius.
- Describe and sketch $|z - 2| = |z + 4i|$.
- Sketch the region satisfying both $|z - 2| \leq 3$ and $\operatorname{Im}(z) \geq 0$.
- Find the greatest and least values of $|z|$ for $|z - 5 - 12i| = 4$.

5. Explain in two sentences why $\arg(z - a) = \theta$ gives a half-line and not a whole line.